THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 36 cm. The length of each tangent is 27 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 1944 cm² **(b)** 486 cm² **(c)** 126 cm² **(d)** 972 cm²

2. Two concentric circles have radii 53 cm and 28 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 45 cm **(b)** 81 cm **(c)** 50 cm **(d)** 90 cm

3. In $\triangle ABC$, right-angled at B, $AB = 16$ cm and $BC = 63$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 7 cm **(b)** 32 cm **(c)** 14 cm **(d)** 8 cm

4. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (6x + 5)$ cm and $QB = (4x + 15)$ cm. The length of QA is: 1

(a) 35 cm **(b)** 37 cm **(c)** 5 cm **(d)** 45 cm

5. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 40^\circ$. The $\angle APB$ is: 1

(a) 80° **(b)** 40° **(c)** 100° **(d)** 50°

6. Two tangents PA and PB are drawn from P and $\angle APB = 90^\circ$. The quadrilateral $OAPB$ is a: 1

(a) rectangle that is not a square **(b)** trapezium
(c) square **(d)** kite that is not a rhombus

7. The length of the tangent from an external point P to a circle is 12 cm and P is 37 cm from the centre. The radius of the circle is: 1

(a) 13 cm **(b)** 35 cm **(c)** 25 cm **(d)** 49 cm

8. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 35 cm. The length of each tangent is 12 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 24 cm **(b)** 94 cm **(c)** 47 cm **(d)** 152 cm

9. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 48 cm. The length of each tangent is 14 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 124 cm² **(b)** 1344 cm² **(c)** 336 cm² **(d)** 672 cm²

10. Two concentric circles have radii 55 cm and 33 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 22 cm **(b)** 88 cm **(c)** 66 cm **(d)** 44 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 72°$. Find the $\angle OAB$. 2

12. A point P is 53 cm from the centre O of a circle of radius 45 cm. Find the length of the tangent drawn from P to the circle. 2

13. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 20°$. Find the $\angle APB$. 2

14. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 84°$. Find the $\angle OAB$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (10x + 3)$ cm and $QB = (6x + 19)$ cm. Find the length of QA. 3

16. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 25$ cm, $BC = 19$ cm and $CD = 16$ cm. Find the length of AD. 3

17. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 35 cm. The length of each tangent is 12 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

OR

Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (12x + 5)$ cm and $QB = (9x + 20)$ cm. Find the length of QA. 3

18. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 30$ cm, $BC = 24$ cm and $CD = 13$ cm. Find the length of AD. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 36 cm. The length of each tangent is 15 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Tangents PA and PB are drawn from a point P at a distance of 55 cm from the centre O of a circle of radius 33 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Prove that in two concentric circles, a chord of the larger circle which touches the smaller circle is bisected at the point of contact. 5

20. Tangents PA and PB are drawn from a point P at a distance of 58 cm from the centre O of a circle of radius 40 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

In $\triangle ABC$, $\angle B = 90°$, $AB = 16$ cm and $BC = 63$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

END OF QUESTION PAPER — Total: 40 Marks